

Wick–Doob terminalization and adapted-resolvent boundary

TECT verification-first repository

claim A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION

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1 Purpose and scope

This note continues R-069 from its displayed finite-cutoff target (6.7). The setting is the hash-pinned six-real production model on the torus of side $L = 16$, at fixed positive density floor, with common entrywise real-even regulation and mutually orthogonal one-shot strict-past controls. The purpose is to decide whether the remaining Gaussian-rooted current can be closed by a conditional Gaussian, paraproduct, or retained-resolvent argument.

The main positive result is an exact Wick–Doob product identity. It combines the GG , CC , and mixed-current channels into the terminal covariance-normal current form while retaining the complete finite-low boundary. The transported covariance trace is already an R-066 summable channel, and the R-069 endpoint-curvature defect is already an R-068 one-use centered form. Thus the covariance and endpoint-curvature side-channel bookkeeping is closed; the full linear-frame model attribution is not.

The result does *not* prove the remaining lower bound. Terminalization lands exactly on the adapted translated-current stochastic coercivity problem which A13 was designed to prove. A second exact identity retains the terminal Schur resolvent and pays its Hilbert–Schmidt trace, but leaves one adapted centered-resolvent chaos. A scalar diagnostic proves that covariance normal ordering does not center this term when its coefficient is adapted. The smallest non-circular successor consequently has two stages: first the balanced non- pp linear-frame model attribution, then a coupled nonlinear rational-frame/cross-square estimate. A7 existence or the desired Nelson moment cannot be cited as either proof.

All identities below are finite-cutoff statements. They use only pointwise conditional independence of a fresh derivative shell from the same-point value variables. They do not identify the target-space heat dummy with a future spatial field and never spatially differentiate it.

2 Strict-past fields and normal current

Let

$$\mathcal{F}_j = \sigma(\omega_k : k \leq j), \quad (a_j, b_{j,i}) \text{ be } \mathcal{F}_{j-1}\text{-measurable.} \quad (2.1)$$

For one terminal shell J , use the coherent value coupling

$$U_J(x) = \sum_{k \leq J} \xi_k(x), \quad (2.2)$$

and define

$$G_{j,i} = \sum_{k < j} d_{k,i}, \quad \Gamma_{j-1,i} = \text{Cov } G_{j,i}, \quad (2.3)$$

$$A_j = A_{<j}, \quad C_{j,i} = D_i A_j, \quad (2.4)$$

$$A_{j+1} = A_j + a_j, \quad C_{j+1,i} = C_{j,i} + b_{j,i}. \quad (2.5)$$

The future heat variable in (2.2) is frozen under D_i ; therefore $G_{j,i}$ contains only exposed spatial derivative shells.

Write the production coefficient in its current frame as

$$B(z) = \sum_{A=1}^3 M_A(z) Q_{\Pi} M_A(z)^T, \quad B_j = B(U_J + A_j), \quad (2.6)$$

and put

$$Q_{j,i} = G_{j,i}^{\otimes 2} - \Gamma_{j-1,i}. \quad (2.7)$$

The covariance-normal current energy at the pre-control endpoint is

$$\mathcal{W}_j = \frac{1}{2} \sum_i \int B_j : [(G_{j,i} + C_{j,i})^{\otimes 2} - \Gamma_{j-1,i}] \, dx. \quad (2.8)$$

At the post-control, pre-noise endpoint define

$$\widehat{\mathcal{W}}_j = \frac{1}{2} \sum_i \int B_{j+1} : [(G_{j,i} + C_{j+1,i})^{\otimes 2} - \Gamma_{j-1,i}] \, dx. \quad (2.9)$$

3 Exact Wick–Doob transport

Fix x and enlarge the strict-past sigma field only by the same-point future values:

$$\mathcal{G}_{j,x} = \mathcal{F}_{j-1} \vee \sigma(\xi_k(x) : k \geq j). \quad (3.1)$$

Common real-even covariance and shell independence give

$$\mathbb{E}[d_{j,i}(x) \mid \mathcal{G}_{j,x}] = 0, \quad \mathbb{E}[d_{j,i}(x)^{\otimes 2} \mid \mathcal{G}_{j,x}] = \Delta \Gamma_{j,i}. \quad (3.2)$$

The fields $B_{j+1}(x)$, $G_{j,i}(x)$, and $C_{j+1,i}(x)$ are $\mathcal{G}_{j,x}$ -measurable. Since

$$G_{j+1,i} = G_{j,i} + d_{j,i}, \quad \Gamma_{j,i} = \Gamma_{j-1,i} + \Delta \Gamma_{j,i}, \quad (3.3)$$

equation (3.2), followed by spatial Fubini, proves

$$\boxed{\mathbb{E} \mathcal{W}_{j+1} = \mathbb{E} \widehat{\mathcal{W}}_j.} \quad (3.4)$$

This is a pointwise conditional statement. It does not assert independence of the complete value and derivative fields.

Polarizing (2.8)–(2.9) gives exactly

$$\widehat{\mathcal{W}}_j - \mathcal{W}_j = \mathcal{R}_j^C + \mathcal{R}_j^\times + \mathcal{R}_j^{G,\text{Wick}}, \quad (3.5)$$

where $\mathcal{R}_j^C$ and $\mathcal{R}_j^\times$ are R-069 (5.7) and (5.9), and

$$\mathcal{R}_j^{G,\text{Wick}} = \frac{1}{2} \sum_i \int (B_{j+1} - B_j) : Q_{j,i} \, dx. \quad (3.6)$$

Summing (3.4)–(3.6) proves the first main identity:

$$\boxed{\mathbb{E} \sum_{j=j_0}^J (\mathcal{R}_j^C + \mathcal{R}_j^\times + \mathcal{R}_j^{G,\text{Wick}}) = \mathbb{E} \mathcal{W}_{J+1} - \mathbb{E} \mathcal{W}_{j_0}.} \quad (3.7)$$

In particular, the GG coefficient alone obeys the Wick–Doob product rule

$$\mathbb{E} \sum_{j=j_0}^J \frac{1}{2} \sum_i \int (B_{j+1} - B_j) : Q_{j,i} \, dx = \frac{1}{2} \mathbb{E} \sum_i \int (B_{J+1} : Q_{J+1,i} - B_{j_0} : Q_{j_0,i}) \, dx. \quad (3.8)$$

Unlike the R-069 scalar-envelope transfer, (3.8) transports the full coefficient increment. Its price is that the result is the intact terminal rough form, not a smooth deterministic multiplier.

At the terminal endpoint $U_J = X_J$, $G_{J+1} = DX_J$, $A_{J+1} = A$, and $C_{J+1} = DA$. Therefore

$$\mathcal{W}_{J+1} = \frac{1}{2} \sum_i \int B(Z_J) : \left[(D_i Z_J)^{\otimes 2} - \Gamma_{J,i} \right] \, dx, \quad (3.9)$$

which is exactly the finite-cutoff A7 covariance-normal current form. The initial term in (3.7) is the *full* finite-low boundary and is retained until the R-066 low-block estimate is applied.

4 Raw terminal-minus-injection form and trace restoration

Let

$$\mathcal{E}_j^{\text{raw}} = \frac{1}{2} \sum_i \int (G_{j,i} + C_{j,i})^T B_j (G_{j,i} + C_{j,i}) \, dx. \quad (4.1)$$

Fresh-noise centering also gives the equivalent raw identity

$$\mathbb{E} \sum_{j=j_0}^J (\mathcal{R}_j^C + \mathcal{R}_j^\times + \mathcal{R}_j^{G,\text{raw}}) = \mathbb{E} \mathcal{E}_{J+1}^{\text{raw}} - \mathbb{E} \mathcal{E}_{j_0}^{\text{raw}} - \frac{1}{2} \mathbb{E} \sum_{j,i} \int \text{Tr}(B_{j+1} \Delta \Gamma_{j,i}) \, dx. \quad (4.2)$$

Thus terminal raw energy and injection may not be estimated separately: at zero control they cancel exactly shell by shell after expectation.

The discrete integration-by-parts identity

$$B_{J+1} \Gamma_J - B_{j_0} \Gamma_{j_0-1} = \sum_{j=j_0}^J \{ (B_{j+1} - B_j) \Gamma_{j-1} + B_{j+1} \Delta \Gamma_j \} \quad (4.3)$$

converts (4.2) into (3.7). Consequently the raw R-069 target (6.7) must not be silently called the Wick terminal form: the previously separated trace channel

$$\frac{1}{2} \sum_{j,i} \int \text{Tr}[(B_{j+1} - B_j) \Gamma_{j-1,i}] \, dx \quad (4.4)$$

must first be restored or charged to its existing budget.

R-066 already proves that (4.4) is a one-use summable channel. For clarity, write $N_j = 2^j$, $a_j = K_j h_j$, and use

$$\|\Gamma_{j-1}\|_{\text{op}} \leq C_\Gamma N_j, \quad \|K_j h_j\|_2 \leq C_K N_j^{-2} \|h_j\|_{\mathcal{H}_j}. \quad (4.5)$$

The frame increment gives a mixed term $C N_j^{-1} \|U_J + A_j\|_2 \|h_j\|$ and a quadratic term $C N_j^{-3} \|h_j\|^2$. Orthogonality and dyadic Cauchy–Schwarz then yield, for a sufficiently large fixed j_0 and every $\eta > 0$,

$$\mathbb{E}|(4.4)| \leq \eta \mathbb{E} \sum_j \|h_j\|_{\mathcal{H}_j}^2 + C_{\eta,j_0}. \quad (4.6)$$

There is an equivalent transported-covariance tail. If $A = A_{J+1}$, put

$$\mathcal{P}_J = \frac{1}{2} \sum_{j,i} \int \text{Tr}\{[B(U_J + A) - B(U_J + A_{j+1})]\Delta\Gamma_{j,i}\} dx. \quad (4.7)$$

Discrete Abel summation gives exactly

$$\mathcal{P}_J = \frac{1}{2} \sum_{k,i} \int \text{Tr}[(B_{k+1} - B_k)\Gamma_{k-1,i}^{\text{hi}}] dx, \quad \Gamma_{k-1}^{\text{hi}} = \sum_{j_0 \leq j < k} \Delta\Gamma_j. \quad (4.8)$$

The same bounds prove

$$|\mathcal{P}_J| \leq C \sum_k \left[N_k^{-1} \|U_J + A_k\|_2 \|h_k\|_{\mathcal{H}_k} + N_k^{-3} \|h_k\|_{\mathcal{H}_k}^2 \right]. \quad (4.9)$$

If $X = \sum_k \|h_k\|^2$ and $W = \sum_k N_k^{-2} \|U_J + A_k\|_2^2$, the dyadic Hardy estimate is

$$W \leq \frac{8}{3} N_{j_0}^{-2} \|U_J\|_2^2 + \frac{2C_K^2}{3} N_{j_0}^{-6} X. \quad (4.10)$$

Hence (4.9) has the same arbitrary control allocation as (4.6), without a sextic allocation. No paraproduct theorem is needed for the covariance tail.

5 Endpoint curvature and exact terminal equivalence

Use the R-069 endpoint lift on the Gaussian derivative only and define

$$\varepsilon_j = \sum_{A,i} \int \langle g_{j,A,i}, Q_{\Pi} E_{j,A,i}^G \rangle dx. \quad (5.1)$$

Then, exactly,

$$\mathcal{R}_j^{G,\text{raw}} = \mathcal{T}_j^G + \mathcal{J}_j^G + \varepsilon_j. \quad (5.2)$$

The absolute form of the global R-069 estimate is

$$|\varepsilon_j| \leq C_* \int |a_j|^2 |G_j|^2 dx, \quad C_* = 24\lambda_Q(\sqrt{34} + 4\sqrt{2}). \quad (5.3)$$

R-068/R-069 transfer the centered scalar part of (5.3) once to the terminal $H^{-1-\kappa}$ model and pay its covariance trace at order N_j^{-3} .

Combining (3.7), (4.4), and (5.2) shows that the correctly normal-ordered R-069 target is

$$\boxed{\mathbb{E}(\mathcal{W}_{J+1} - \mathcal{W}_{j_0}) - \mathbb{E} \sum_{j=j_0}^J \varepsilon_j.} \quad (5.4)$$

Equivalently, the literal raw target (6.7) is (5.4) plus the trace channel (4.4). Since (4.4) and (5.3) already have arbitrary one-use allocations, the desired bound for (6.7) is equivalent, up to already proved budgets, to

$$\mathbb{E}\mathcal{W}_{J+1} \geq -\eta \mathbb{E} \sum_j \|h_j\|_{\mathcal{H}_j}^2 - \zeta \mathbb{E} \|Z_J\|_6^6 - C_{\eta,\zeta,j_0}. \quad (5.5)$$

Equation (5.5) is the terminal translated-current stochastic coercivity problem itself. Thus Wick–Doob transport closes the shell bookkeeping but does not provide a new lower-bound mechanism.

6 Exact retained terminal resolvent

There is a second useful exact reduction. Whiten the terminal Gaussian as ξ , write its Cameron–Martin displacement as $\hat{h}$, and, at the actual endpoint $z = K_J(\xi + \hat{h})$, define

$$L_z = M_{B(z)}^{1/2} DK_J, \quad T_z = L_z^* L_z \succeq 0. \quad (6.1)$$

For $p > 0$ put

$$R_z = (I + pT_z)^{-1}, \quad S_z = T_z R_z. \quad (6.2)$$

Although T_z is random, the following is a pointwise Hilbert-space completion and uses no independence:

$$\begin{aligned} V_J^{\text{ren}}(z) + \frac{1}{2p} \|\hat{h}\|^2 &= \frac{1}{2} \{ \langle \xi, S_z \xi \rangle - \text{Tr } S_z \} - \frac{p}{2} \text{Tr}(T_z^2 R_z) \\ &\quad + \frac{1}{2} \|(T_z + p^{-1}I)^{1/2}(\hat{h} + pR_z T_z \xi)\|^2. \end{aligned} \quad (6.3)$$

To verify the trace term, use $S_z = T_z - pT_z^2 R_z$. To verify the square, expand it and use $(T_z + p^{-1}I)R_z = p^{-1}I$.

The deterministic trace loss closes:

$$\text{Tr}(T_z^2 R_z) \leq \|T_z\|_{\mathfrak{S}_2}^2 \leq C \|B(z)\|_2^2 \leq C \|z\|_4^4. \quad (6.4)$$

Consequently, for every $\zeta > 0$,

$$\frac{p}{2} \text{Tr}(T_z^2 R_z) \leq \zeta \|z\|_6^6 + C_{p,\zeta}. \quad (6.5)$$

The sole unresolved term in (6.3) is therefore

$$\boxed{\frac{1}{2} \mathbb{E} \{ \langle \xi, S_{Z_J} \xi \rangle - \text{Tr } S_{Z_J} \}}. \quad (6.6)$$

If S were deterministic or independent of ξ , (6.6) would vanish and the A9 frozen-shell theorem would close the form. At the adapted terminal endpoint it is neither.

7 Adapted centering and Stein no-go diagnostics

The failure of automatic centering is already visible in one Gaussian coordinate. Let $\xi \sim \mathcal{N}(0, 1)$ and choose the bounded adapted coefficient

$$z(\xi) = A \mathbf{1}_{\{|\xi| \leq a\}}. \quad (7.1)$$

Smooth approximations give the same limiting conclusion. Since

$$\int_{-a}^a (x^2 - 1) \varphi(x) dx = -2a\varphi(a), \quad (7.2)$$

one has

$$\mathbb{E} \frac{1}{2} z(\xi)^2 (\xi^2 - 1) = -a\varphi(a)A^2 < 0. \quad (7.3)$$

For the scalar resolvent $S = z^2/(1 + pz^2)$ the same calculation gives

$$\mathbb{E} \frac{1}{2} S(\xi) (\xi^2 - 1) = -\frac{a\varphi(a)A^2}{1 + pA^2} < 0. \quad (7.4)$$

This is a route diagnostic, not a production counterexample: realizing (7.1) as a spatial Cameron–Martin control has a cost which the diagnostic does not model. It proves only that covariance normal ordering and a terminal resolvent do not themselves center an adapted coefficient.

Direct Gaussian integration by parts also leaves an unauthorized norm. If $G = L\omega$, $U = K\omega$, and the same-point evenness gives $LK^* = 0$, then for an adapted shift $A(\omega)$,

$$\mathbb{E}[G^{\otimes 2} : F(U + A)] = \mathbb{E} \operatorname{Tr}\{L^* D_\omega^2 [F(U + A)] L\}. \quad (7.5)$$

The right side contains

$$D^2 F[DA, DA] + DF[D^2 A], \quad (7.6)$$

and the mixed current similarly contains DA and DC . Strict-past adaptation kills only the same-shell diagonal derivative; a later control may still differentiate an earlier innovation. For example

$$h_2^{(n)}(\omega_1) = \sin(n\omega_1) \quad (7.7)$$

has uniformly bounded L^2 control energy but Malliavin derivative norm of order n . The declared Boue–Dupuis cost controls no such derivative. Therefore a pure Stein proof would impose an additional hypothesis and would not extend to the stated finite-energy drift class.

Finally, a global entropy shortcut is circular. Applying the Gibbs variational formula directly to (5.5) bounds it by $-\log \mathbb{E} \exp(-pV_J^{\text{ren}}(X_J))$, which is precisely the open Nelson moment. A7 supplies existence and convergence of the composite, not this negative exponential bound.

8 The exact non-circular successor

Write $U = X_J$, $A = Z_J - X_J$, $G_i = D_i U$, $C_i = D_i A$, and

$$M_{r,0} = M_r(U), \quad M_{r,1} = M_r(U + A), \quad \Delta M_r = M_{r,1} - M_{r,0}, \quad r = 1, 2, 3. \quad (8.1)$$

The terminal translated form is exactly

$$\begin{aligned} \Delta V_J^{\text{ren}}(U; A) &= \frac{1}{2} \langle B(U + A) - B(U), Q_J \rangle \\ &\quad + \sum_{r,i} \langle M_{r,1}^T G_i, Q_{\Pi} M_{r,1}^T C_i \rangle + \frac{1}{2} \sum_{r,i} |M_{r,1}^T C_i|_{Q_{\Pi}}^2, \end{aligned} \quad (8.2)$$

where $Q_J = \sum_i (G_i^{\otimes 2} - \Gamma_{J,i})$.

Subtract the uncontrolled conservative first variation

$$\mathcal{S}_0(U; A) = \frac{1}{2} \langle DB(U)[A], Q_J \rangle + \sum_{r,i} \langle M_{r,0}^T G_i, Q_{\Pi} M_{r,0}^T C_i \rangle. \quad (8.3)$$

Then ordinary algebra gives

$$\begin{aligned} \Delta V_J^{\text{ren}} - \mathcal{S}_0 &= \frac{1}{2} \langle B(U + A) - B(U) - DB(U)[A], Q_J \rangle \\ &\quad + \sum_{r,i} \{ \langle M_{r,0}^T G_i, Q_{\Pi} (\Delta M_r)^T C_i \rangle \\ &\quad + \langle (\Delta M_r)^T G_i, Q_{\Pi} M_{r,1}^T C_i \rangle + \frac{1}{2} |M_{r,1}^T C_i|_{Q_{\Pi}}^2 \}. \end{aligned} \quad (8.4)$$

The exact surviving transported-current paraprodut is

$$\boxed{\mathcal{C}_J = \sum_{r,i} \langle M_{r,0}^T G_i, Q_{\Pi} (\Delta M_r)^T C_i \rangle.} \quad (8.5)$$

The fundamental theorem of calculus must retain its linear part:

$$\Delta M_r = DM_r(U)[A] + E_r^M, \quad E_r^M = \int_0^1 [DM_r(U + tA) - DM_r(U)][A] dt,$$

and hence $\mathcal{C}_J = \mathcal{L}_J + \mathcal{N}_J$, where

$$\mathcal{L}_J = \sum_{r,i} \langle M_{r,0}^T G_i, Q_{\Pi} DM_r(U)[A]^T C_i \rangle, \quad \mathcal{N}_J = \sum_{r,i} \langle M_{r,0}^T G_i, Q_{\Pi} (E_r^M)^T C_i \rangle. \quad (8.6)$$

Write $Q_{\Pi} = (q_{mn})_{m,n=1}^2$ and $M_r = [p_r, v_r]$. Put

$$P_{r,i} = p_r(U) \cdot G_i, \quad V_{r,i} = v_r(U) \cdot G_i,$$

and, with $\rho = |U|^2$, $m_r = U^T S_r U$,

$$K_{r,p} = 2S_r, \quad K_{r,v} = 2(S_r - q_r I) - \frac{v_r U^T + U v_r^T}{\rho + e}, \quad d\omega_r = \frac{d\rho \wedge dm_r}{\rho + e}.$$

Direct differentiation gives the exact symmetric–Cartan split

$$Dp_r(U)[A] \cdot C_i = A^T K_{r,p} C_i, \quad Dv_r(U)[A] \cdot C_i = A^T K_{r,v} C_i + \frac{1}{2} d\omega_r(U)[A, C_i].$$

Consequently

$$\mathcal{L}_J = \mathcal{L}_J^{\text{sym}} + \mathcal{L}_J^{\text{Car}},$$

where

$$\begin{aligned} \mathcal{L}_J^{\text{sym}} = \sum_{r,i} \int \{ & q_{11} P_{r,i} A^T K_{r,p} C_i \\ & + q_{12} [P_{r,i} A^T K_{r,v} C_i + V_{r,i} A^T K_{r,p} C_i] \\ & + q_{22} V_{r,i} A^T K_{r,v} C_i \} dx, \end{aligned}$$

and

$$\mathcal{L}_J^{\text{Car}} = \frac{1}{2} \sum_{r,i} \int (q_{12} P_{r,i} + q_{22} V_{r,i}) d\omega_r(U)[A, C_i] dx.$$

Since $K_{r,p}$ and $K_{r,v}$ are symmetric, torus integration by parts gives

$$\mathcal{L}_J^{\text{sym}} = \sum_r \langle \mathbb{Q}_r^{\text{lin}}(U), A^{\otimes 2} \rangle,$$

$$\mathbb{Q}_r^{\text{lin}} = -\frac{1}{2} \sum_i \partial_i [q_{11} P_{r,i} K_{r,p} + q_{12} (P_{r,i} K_{r,v} + V_{r,i} K_{r,p}) + q_{22} V_{r,i} K_{r,v}].$$

The pure p – p subchannel is therefore

$$\begin{aligned} \mathcal{L}_J^{pp} &= q_{11} \sum_{r,i} \int \partial_i (U^T S_r U) \partial_i (A^T S_r A) dx \\ &= -q_{11} \sum_r \langle \Delta(U^T S_r U), A^T S_r A \rangle. \end{aligned} \quad (8.7)$$

After covariance centering, R-068 pays (8.7) once. The production weight q_{11} and the sum over all generators are essential. The same payment for all of $\mathcal{L}_J^{\text{sym}}$ additionally requires the balanced rational-current model lemma $\mathbb{Q}_r^{\text{lin}} \in L^p(\Omega; H^{-1-\kappa})$; R-070 has not yet proved or pinned that lemma for the non- pp terms.

The exact $\mathcal{L}_J^{\text{Car}}$ is a Gaussian-current pairing with $A \wedge DA$. If $X = \|A\|_{H^2}^2$, $Y = \|A\|_6^6$, and $\theta = (1 + 2\kappa)/3$, interpolation gives

$$\|A \wedge DA\|_{H^\kappa} \leq CX^{(1+\theta)/2}Y^{(1-\theta)/6}.$$

At $\kappa = 1/10$, the powers are $X^{7/10}Y^{1/10}$, and weighted Young gives

$$|\langle \mathbb{J}, A \wedge DA \rangle| \leq \eta X + \zeta Y + C\eta^{-7/2}\zeta^{-1/2}\|\mathbb{J}\|_{H^{-1/10}}^5. \quad (8.8)$$

The balanced Gaussian-current model has the required fifth moment, so (8.8) pays $\mathcal{L}_J^{\text{Car}}$. This does not prove the missing $H^{-1-\kappa}$ model attribution for the non- pp part of $\mathcal{L}_J^{\text{sym}}$.

With $d_{r,i} = (\Delta M_r)^T G_i$ and $c_{r,i} = M_{r,1}^T C_i$, define

$$\begin{aligned} \mathcal{R}_{\text{shift}} = & \mathcal{N}_J + \sum_{r,i} \{ \langle d_{r,i}, Q_{\text{II}} c_{r,i} \rangle + \frac{1}{2} |c_{r,i}|_{Q_{\text{II}}}^2 \} \\ & + \frac{1}{2} \langle B(U + A) - B(U) - DB(U)[A], Q_J \rangle. \end{aligned} \quad (8.9)$$

The exact identity is

$$\Delta V_J^{\text{ren}} - \mathcal{S}_0 = \mathcal{L}_J^{\text{sym}} + \mathcal{L}_J^{\text{Car}} + \mathcal{R}_{\text{shift}}.$$

The terms in (8.9) must remain coupled. Separating them recreates the old saturated rational-frame paraproduct, while completing the cross/square globally creates $|A|^2|DU|^2$ with terminal covariance and destroys the shellwise N_J^{-3} trace gain. The smallest honest successor is therefore:

Successor (8.10). First prove the balanced model attribution for the non- pp part of $\mathbb{Q}_r^{\text{lin}}$ and pay $\mathcal{L}_J^{\text{sym}}$ once through R-068. Then prove a full Q_{II} -weighted predictable rational-frame/cross-square inequality for (8.9), retaining $\mathcal{N}_J$, the coefficient remainder, and terminal square jointly and using no Malliavin derivative of the control.

Only after (8.10) is closed can localization and lower semicontinuity extend the result to all declared finite-energy drifts, followed by controlled-shell one-use and the $q = 10/9$ Nelson theorem.

9 Devil's-advocate review

1. **“The full coefficient Wick product may be transported to the terminal shell.” DISMISSED as an objection.** Equations (3.1)–(3.8) prove this exactly, using only same-point fresh-noise centering. The terminal rough form and full finite-low boundary are retained.
2. **“The raw R-069 expression already equals the Wick terminal form.” VALID WITH MITIGATION.** The statement is true only after restoring the trace channel (4.4). R-066 proves that channel summable, but it may not be silently omitted from the identity.
3. **“Terminalization proves (6.7).” UPHELD as false.** Equations (5.4)–(5.5) show that it lands on the terminal translated-current coercivity theorem itself. Using the desired Nelson moment there is circular.
4. **“The retained resolvent centers the terminal quadratic.” UPHELD as false.** Equations (7.1)–(7.4) give a bounded scalar adapted coefficient for which the centered resolvent expectation is strictly negative. This is a route no-go, not a production counterexample.
5. **“Stein integration uses only the control energy.” UPHELD as false.** Equations (7.5)–(7.7) exhibit uncontrolled first and second Malliavin derivatives of later controls.

6. **“The covariance tail needs a new paraprodut theorem.” DISMISSED.** Abel summation, R-066 frame bounds, orthogonal ranges, and the dyadic Hardy estimate (4.10) give an arbitrary one-use allocation.
7. **“The pure p - p integration-by-parts channel has unit weight and exhausts the linear frame.” UPHELD as false.** Equation (8.7) carries the production factor q_{11} and the generator sum. The q_{12}/q_{22} rational-frame terms remain in $\mathbb{Q}_r^{\text{lin}}$ and require the stated balanced-model attribution.
8. **“The nonlinear integral in (8.6) equals ΔM_r .” UPHELD as false.** It equals $\Delta M_r - DM_r(U)[A]$. Dropping the linear term loses $\mathcal{L}_J$; equations (8.6)–(8.9) retain it exactly.
9. **“Termwise absolute bounds on (8.4) are sufficient.” UPHELD as false.** They destroy the coupled $\langle d, c \rangle + |c|^2/2$ structure and recreate the previously registered critical saturation. The next estimate must retain the positive square.
10. **“This result proves finite-energy one-use or Nelson.” UPHELD as an overclaim.** It proves an exact finite-cutoff reduction for regular one-shot controls and sharpens the successor. The full weighted rational-frame/cross-square residual, finite-energy extension, one-use, Nelson theorem, interacting measure, floor removal, and infinite volume remain open.

Executed code-discipline audit. Signs, q_{11} /half-factors, regulator/covariance conventions, and Sobolev orders are independently asserted. All quantities use the nondimensional A1 normalization. Same-fixture coarse/fine quadrature and Fourier envelopes test convergence; every diagnostic is recomputed rather than pasted, with zero- control, deterministic, positive-floor, and adapted-scalar limit cases. These are regression checks, not a proof of (8.10). Before child execution the integrated verifier also hash-checks the pinned R-069 helper and its complete NPC/translation/UV runtime dependency closure, including the translation and strict-past manifests read by the NPC helper.

10 Reproduction

The fail-closed verifier first checks R-070 source/PDF hashes and the transitive R-069 runtime dependency closure. Only then does it run the primary, non-importing independent, prior-evidence, and repository-integration gates:

```
python codes/foundations/a13_classii_wick_doob_terminal_resolvernt_reduction_verify.py
```

The primary route checks polarization, the full weighted frame-linearization split, Wick/raw telescopes, traces, Hardy, Schur completion, and adapted centering. Both routes report same-fixture coarse/fine envelopes. Neither executable claims to prove (8.10).

Result footer

Result ID: A13-CLASSII-WICK-DOOB-TERMINAL-RESOLVENT-REDUCTION (R-070)

Precise statement:	The full coefficient Wick current telescopes into the terminal covariance-normal current; raw/Wick trace and transported tail are paid; exact terminal Schur completion isolates one adapted resolver chaos; the full linear frame is decomposed without deletion.
Scope:	Fixed finite cutoff; regular mutually orthogonal strict-past one-shot controls; positive density floor.
Dependencies:	A1 production functional; R-066; R-068; R-069.
Evidence grade:	ANALYTIC, EXACT, EXECUTED.
Reproduction command:	python codes/foundations/

	a13_classii_wick_doob_terminal_resolvent_reduction_verify.py
Expected output:	primary 22/22; independent 16/16; integrated 47/47; aggregate 85/85; exit 0.
Falsification gate:	A conditional moment, telescope, q11/frame split, trace, Hardy, Schur, hash, PDF QA, or assertion fails.
Tier before / after:	T4 / T4; no promotion.
No-overclaim statement:	The non-pp symmetric-model attribution, shifted rational-frame/cross-square bound, finite-energy, one-use, Nelson, and all removal limits remain open.
Next required action:	Prove the balanced linear-frame model lemma and then the coupled residual (8.9)--(8.10).

Final boundary. This result terminalizes R-069 (6.7), proves that its covariance and endpoint curvature side channels are already one-use, and isolates one adapted centered-resolvent/transported-current problem. It does not prove the non-*pp* linear-frame model attribution, the coupled nonlinear lower bound, or any downstream constructive claim.